

Pedram Navard

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About Me

I work at the intersection of **computer graphics**, **computer vision**, and **generative AI**, with an interest in turning generative-AI outputs into playable, interactive experiences rather than static images. This drove my BSc thesis that makes AI-generated content explorable in 3D.

Education

BSc in Computer Science and Engineering

Sept 2022 – Feb 2026

Islamic Azad University, Gilan Branch, Iran

GPA: 3.1 / 4.0

- Thesis: *Interactive 3D Visualization of Generative-AI Outputs: A Real-Time Web Graphics Pipeline*
 - Built a browser-based rendering pipeline (Three.js / WebGL) that displays AI-generated images and assets as interactive, real-time 3D scenes.
 - Converted 2D generative outputs into navigable 3D using depth estimation and image-based scene construction.
 - Added interactive camera and scene controls so users explore generated content rather than view it as static images.
- Advisor: Dr. Hossein Azgomi, Google Scholar
- Relevant coursework: Linear Algebra, Calculus, Probability & Statistics, Data Structures & Algorithms, Computer Graphics, Digital Image Processing, Artificial Intelligence, Databases.

Projects

COMPUTER VISION

Real-Time Video Object Detection and Segmentation Pipeline ·

2023 – 2024

- Developed a computer vision pipeline integrating YOLOv8 object detection with the Segment Anything Model (SAM) for real-time person segmentation.
- Combined bounding-box detection with prompt-based instance segmentation to generate accurate pixel-level masks and overlay visualizations.
- Automated end-to-end video processing, including inference, visualization, and MP4 output generation for side-by-side performance comparison.

Monocular Depth Estimation using latent diffusion models ·

2024 – 2025

- implementing a computer vision pipeline from the Marigold diffusion model paper for monocular depth estimation
- Integrated pretrained model checkpoints, inference scripts, and image preprocessing pipelines to perform depth estimation on custom input images.

COMPUTER GRAPHICS

Interactive 3D Scene Reconstruction from Single Images ·

2024 – 2025

- Implemented a single-image 3D scene reconstruction pipeline based on the *3D Photo Inpainting* framework to generate novel viewpoints from RGB images.
- Converted monocular depth maps into layered 3D scene representations using depth-based warping and image inpainting techniques.
- Built an interactive WebGL/Three.js viewer for real-time camera navigation and visualization of reconstructed 3D scenes.

- Implemented a human reconstruction pipeline based on the *PIFuHD* framework to recover detailed 3D human geometry from single RGB images.
- Integrated image preprocessing, silhouette extraction, and neural implicit surface reconstruction to generate textured 3D human meshes.
- Visualized reconstructed meshes in an interactive Three.js viewer with real-time camera controls and lighting.

Technical Skills

Programming: Python, JavaScript, SQL, PHP

ML / Vision: scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Graphics: *add the tools you actually use: Three.js, WebGL, OpenGL, GLSL, Blender*

Foundations: linear algebra, probability, and optimization applied to perception and modeling

Web / Tools: React.js, Next.js, Firebase, Redux Toolkit, GSAP, Git

Honors & Activities

- Built a real-time polling application at the IranWebFest Hackathon (React/Node.js).
- Solved 100+ algorithmic problems on LeetCode (arrays, dynamic programming).
- Author of technical articles on React best practices (Medium).
- Open-source contributor; interests include applied ML/AI and creative developer tools (e.g., ComfyUI extensions).

Languages & Test Scores

Persian (native) · English — **TOEFL iBT 96/120** (Speaking 25)

References

Dr. Hossein Azgomi — Assistant Professor, Software Engineering, Islamic Azad University
hossein.azgomi@gmail.com (thesis advisor)

Dr. Azam Andalib — Assistant Professor, Software Engineering, Islamic Azad University
azam.andalib@gmail.com